UE23CS342BA1: SCME Project

PES UNIVERSITY

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B. Tech. CSE — 6th Semester | Jan – May 2026
UE23CS342BA1
SUPPLY CHAIN MANAGEMENT (SCM) FOR ENGINEERS

PROJECT REPORT

on

Problem No. P5

SupplyNex Pvt Ltd — Centralized SCM Dashboard with ARIMA Demand Forecasting

Prepared On Date: April 20, 2026

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1. Introduction & Problem Statement Analysis

1.1 Problem Statement

Companies operating large-scale supply chains often lack a unified, centralized monitoring system. This absence forces managers to rely on fragmented data from multiple silos, leading to inefficient decision-making, inventory stockouts, excess stock, poor visibility across warehouses and procurement channels, elevated operational costs, and missed revenue opportunities.

Problem Statement P5 — Centralized Monitoring — directly addresses these challenges by requiring the design of a comprehensive, data-driven monitoring ecosystem that consolidates all supply chain operations into a single decision-support platform.

1.2 Scope of the Solution

SupplyNex Pvt Ltd is an imaginary supply chain company operating across multiple regions in India, managing a diverse product portfolio spanning Electronics, Raw Materials, FMCG, and Packaging. The scope of this project covers:

- Centralized inventory and stock monitoring across all warehouses
- Real-time order and delivery tracking with status classification
- Procurement cost and supplier performance analytics
- ML-driven demand forecasting (ARIMA) for the next 6 months
- Interactive KPI dashboard for executive-level decision-making

1.3 SCM Area Addressed

This project addresses all five core SCM pillars simultaneously: Inventory Management, Order & Delivery Visibility, Procurement Reliability, Supplier Performance, and Demand Forecasting. The centralized nature of the solution ensures that supply chain managers have a single pane of glass across all operations.

1.4 Proposed Solution Approach

Our solution follows a three-layer architecture:

- Data Layer: A relational database (10 normalized tables) populated with realistic sample data covering 16 months of historical transactions.
- ML Layer: ARIMA time-series forecasting models trained per product to predict the next 6 months of demand, with automated model selection via AIC grid search.
- Presentation Layer: An interactive HTML dashboard featuring live KPIs, Chart.js visualizations, region/category filters, and warehouse stock alert tables.

2. Data Structure & Database Design

2.1 Entity-Relationship Overview

The SupplyNex database consists of 10 interrelated tables designed to capture every relevant dimension of supply chain operations. The schema follows Third Normal Form (3NF) to eliminate data redundancy and ensure referential integrity.

2.2 Table Descriptions

Table	Primary Key	Purpose
COMPANY	company_id	Top-level entity — regional SCM companies
WAREHOUSE	warehouse_id	Warehouses linked to companies with capacity
PRODUCT	product_id	Product catalog with pricing and categories
INVENTORY	inventory_id	Stock levels per product per warehouse
SUPPLIER	supplier_id	Vendor list with performance scores
PROCUREMENT_ORDER	proc_order_id	Procurement requests per supplier/product
ORDERS	order_id	Customer/internal orders per warehouse

ORDER_ITEMS	item_id	Line-items linking orders to products
DELIVERY	delivery_id	Shipment tracking per order
DEMAND_FORECAST	forecast_id	ARIMA model outputs per product/warehouse

2.3 Key Relationships

- COMPANY → WAREHOUSE (1:M) — A company operates multiple warehouses
- WAREHOUSE → INVENTORY (1:M) — Each warehouse tracks multiple product stock levels
- PRODUCT → INVENTORY, ORDER_ITEMS, PROCUREMENT_ORDER, DEMAND_FORECAST (1:M) — Product is the central entity
- ORDERS → ORDER_ITEMS (1:M) — Orders decompose into line-items
- ORDERS → DELIVERY (1:1) — Each order has exactly one delivery record
- SUPPLIER → PROCUREMENT_ORDER (1:M) — Suppliers fulfill procurement requests

2.4 Sample Dataset Summary

Entity	Records	Date Coverage
Companies	5	N/A
Warehouses	10	N/A
Products	15	N/A
Inventory Records	20+	Jan 2025 – Apr 2026
Orders	25+	Jan 2025 – Apr 2026
Procurement Orders	20+	Jan 2025 – Apr 2026
Demand Forecast Rows	30 (ML output)	May 2026 – Oct 2026

3. Dashboard Design & Development

3.1 Technology Stack

- Frontend: HTML5, CSS3, JavaScript (ES6)
- Charting Library: Chart.js v4.4.1
- Deployment: Self-contained single-file HTML — runs offline in any browser
- Interactivity: Region and Category filters with dynamic KPI recalculation

3.2 KPI Metrics

The dashboard presents six executive-level KPIs at the top of the interface, each with a delta indicator showing trend direction:

KPI	Value	Significance
Total Orders	4,821	Overall order volume; +12% MoM growth
Inventory Turnover	6.4x	Efficiency of stock conversion (+0.8x)
Stockout Rate	3.2%	Alerts for replenishment action needed
On-Time Delivery	91%	Service level; +4% vs Q4 2025
Procurement Cost	₹2.1 Cr	Budget monitoring; 7% overspend flag
Supplier Score	78/100	Vendor health composite index

3.3 Visualizations

- Inventory Levels Bar Chart — Current stock vs. reorder point per category
- Order Status Doughnut Chart — Fulfilled (58%), In Transit (22%), Pending (14%), Delayed (6%)

- Procurement Cost vs Revenue Line Chart — 16-month trend in ₹ Lakhs
- ML Demand Forecast Chart — Actual Jan–Apr 2026 + ARIMA forecast May–Oct 2026
- Supplier Performance Horizontal Bar Chart — Top 5 vendors scored out of 100
- Warehouse Stock Alert Table — Real-time status (OK / Low / Stockout Risk) per item

3.4 Interactive Elements

- Region filter (All / North / South / West / East) — adjusts KPI values dynamically
- Category filter (All / Electronics / Raw Materials / FMCG) — for drill-down analysis
- Fix Stockout button — contextually triggers an action prompt for Raw Materials

4. Machine Learning Integration — ARIMA Demand Forecasting

4.1 Why ARIMA?

ARIMA (AutoRegressive Integrated Moving Average) was selected as the forecasting model because our demand data exhibits non-stationarity (clear upward trends) without seasonality. ARIMA is well-suited to supply chain demand forecasting as it:

- (a) handles non-stationary data via the integration parameter d
- (b) captures autocorrelation patterns in historical demand
- (c) produces probabilistic confidence intervals useful for inventory buffer planning
- (d) is interpretable and accepted in industry practice

4.2 Data Used for Training

Historical monthly demand data was sourced from the DEMAND_FORECAST table in the SupplyNex sample database, covering January 2025 to April 2026 (16 months) across 5 representative products: Laptop Pro X (Electronics), Steel Sheets (Raw Materials), Copper Wire (Raw Materials), Rice 5kg Pack (FMCG), and Cardboard Box (Packaging).

4.3 Methodology

Step 1 — Stationarity Testing (ADF Test)

The Augmented Dickey-Fuller (ADF) test was applied to each training series. All 5 products returned p-values > 0.05, indicating non-stationary data. This confirmed that differencing (parameter $d \geq 1$) was required before modeling.

Step 2 — Automated Model Selection (Grid Search)

A grid search was performed over ARIMA(p, d, q) parameter combinations with $p \in \{0..3\}$, $d \in \{0..2\}$, $q \in \{0..3\}$. The model with the lowest AIC (Akaike Information Criterion) score was selected per product, balancing fit quality against model complexity.

Step 3 — Train/Validation Split

13 months (Jan 2025 – Jan 2026) were used for training, with the remaining 3 months (Feb–Apr 2026) held out for validation. Three accuracy metrics were computed: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE).

Step 4 — Full Refit and Forecasting

After validation, each model was refit on the full 16-month dataset and used to produce a 6-month forecast (May–October 2026) with 95% confidence intervals.

4.4 Results

Product	Best ARIMA	MAE	RMSE	MAPE	Grade
Laptop Pro X	(2,2,0)	11.80	12.52	5.33%	**Excellent**
Steel Sheets	(2,2,1)	14.86	15.03	3.77%	**Excellent**
Copper Wire	(0,2,2)	6.51	8.27	2.58%	**Excellent**
Rice 5kg Pack	(0,2,2)	7.60	7.71	1.25%	**Outstanding**
Cardboard Box	(0,2,3)	5.20	6.41	2.59%	**Excellent**

All MAPE values are below 6%, well within the industry-accepted threshold of 10% for 'good' forecasting accuracy. Rice 5kg Pack achieved the best MAPE of 1.25%, reflecting the highly stable and predictable demand patterns for FMCG goods.

4.5 Forecast Output (May – October 2026)

Product	May	Jun	Jul	Aug	Sep	Oct
Laptop Pro X	230	235	240	245	250	255
Steel Sheets	401	410	415	418	425	431
Copper Wire	264	268	273	277	281	286
Rice 5kg Pack	629	638	647	656	666	675
Cardboard Box	209	212	216	220	223	227

4.6 ARIMA Forecast Plots

The figure below shows the individual forecast charts for all 5 products along with the Model Accuracy (MAPE) bar chart. Each chart shows actual historical demand (green), validation overlay (orange dashed), ARIMA forecast (purple dashed), and 95% confidence intervals (shaded region).

Figure 1: ARIMA Demand Forecasting Results — All 5 Products + MAPE Accuracy Summary

4.7 Business Impact of ML Integration

- Inventory Optimization: Forecast values directly feed reorder point calculations, reducing stockout and overstock risk
- Procurement Planning: Procurement orders can be placed 2–3 months ahead based on forecasted demand trends
- Budget Forecasting: Finance teams can model future cost scenarios using the 6-month demand pipeline
- Automated Alerts: When forecasted demand exceeds current stock + in-transit procurement, an alert is triggered automatically

5. Key Insights Derived

5.1 Inventory

- Raw Material B has a critical stockout risk (110 units vs. 300 reorder point) and requires immediate replenishment
- Raw Material A is also at risk (420 units vs. 600 reorder point) — procurement lead times must be accelerated
- Electronics inventory is healthy at 1,200 units vs. 500 reorder point

5.2 Orders & Delivery

- 91% on-time delivery rate is strong but leaves room for improvement — delayed orders (6%) are concentrated in the West region
- Order volumes have grown 12% month-over-month, signaling the need for warehouse capacity expansion planning

5.3 Procurement & Cost

- Procurement cost has risen 7% above budget (₹2.1 Cr actual vs. ₹1.96 Cr target), driven by raw material price volatility
- VendorD (score: 65) and VendorE (score: 55) are underperforming — renegotiation or replacement is recommended

5.4 Demand Trends

- All 5 product categories show consistent upward demand trends with CAGR of approximately 8–12% annually
- FMCG demand (Rice 5kg Pack) is the most predictable (MAPE 1.25%) and should be prioritized for automated replenishment
- Electronics demand (Laptop Pro X, MAPE 5.33%) has the highest forecast uncertainty — larger safety stock buffer recommended

6. Challenges & Learnings

6.1 Challenges Faced

- ARIMA Parameter Tuning: With only 16 months of historical data, some ARIMA parameter combinations produced convergence warnings. This was resolved by relying on AIC-guided model selection rather than manual tuning.
- Data Realism: Designing synthetic data that mimics real-world supply chain variability (demand spikes, stockouts, supplier delays) while maintaining internal consistency required careful planning.
- Dashboard Responsiveness: Ensuring the Chart.js dashboard rendered correctly across mobile and desktop screen sizes required CSS media query adjustments.

6.2 Learnings

- Time-series forecasting quality is highly sensitive to data quantity — more historical months would yield lower MAPE
- Centralized dashboards dramatically reduce the cognitive load on supply chain managers by surfacing only actionable exceptions
- ARIMA works well for trend-dominant, non-seasonal data, but seasonal products would benefit from SARIMA or Prophet models

6.3 Future Improvements

- Integrate SARIMA or Facebook Prophet for products with seasonal demand patterns
- Add anomaly detection (Isolation Forest) to flag unusual procurement or delivery events
- Connect the dashboard to a live database via REST API for real-time updates
- Add user role management (executive view vs. warehouse manager view) with different KPI granularities
- Implement automated email/SMS alerts when stock crosses reorder thresholds

7. Deliverables Summary

Deliverable	File	Status
ER Diagram	p5_scm_er_diagram.html	Complete — 10 tables, all FK relationships
Sample Dataset (SQL)	scm_p5_sample_data.sql	Complete — all 10 tables populated
Interactive Dashboard	p5_scm_centralized_monitoring_dashboard.html	Complete — KPIs, 5 charts, filters
ML Code (ARIMA)	p5_scm_arima_forecast.py	Complete — ADF + grid search + forecast
ML Forecast Output (CSV)	p5_arima_forecast_output.csv	Complete — 30 rows, 5 products × 6 months
ML Forecast Plots	p5_arima_forecast_plots.png	Complete — 6 charts, MAPE accuracy bar
Project Report	SCM23CS_P5_497-561-564.docx	Complete — this document

8. References

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Github Repo : https://github.com/SAGARSINGH-nbmr/SCM_P5_497_561_564